

Untitled Driving Game

THESIS | DEVELOPMENT

Alexander Freyr | Alexander Follin | Aleksandra Pavlova-Verevkina

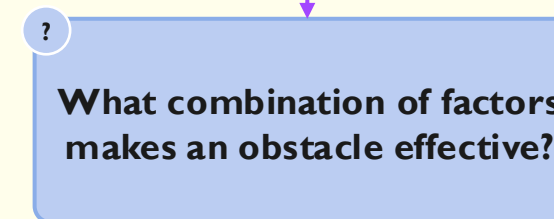
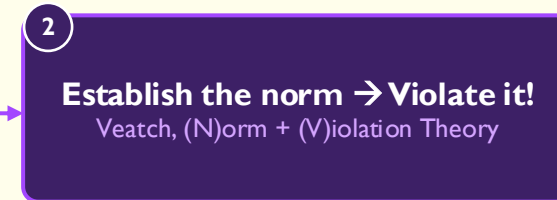
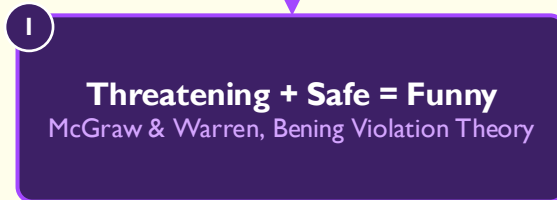
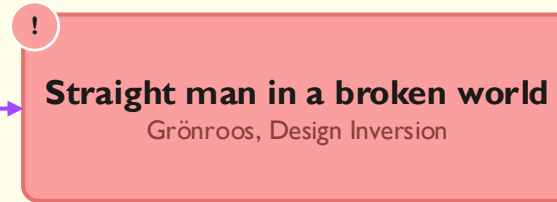
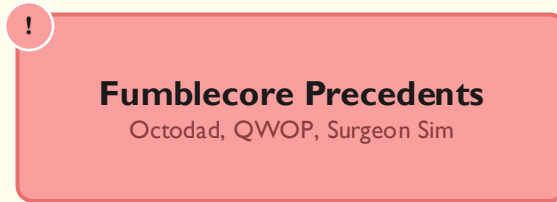


Research context

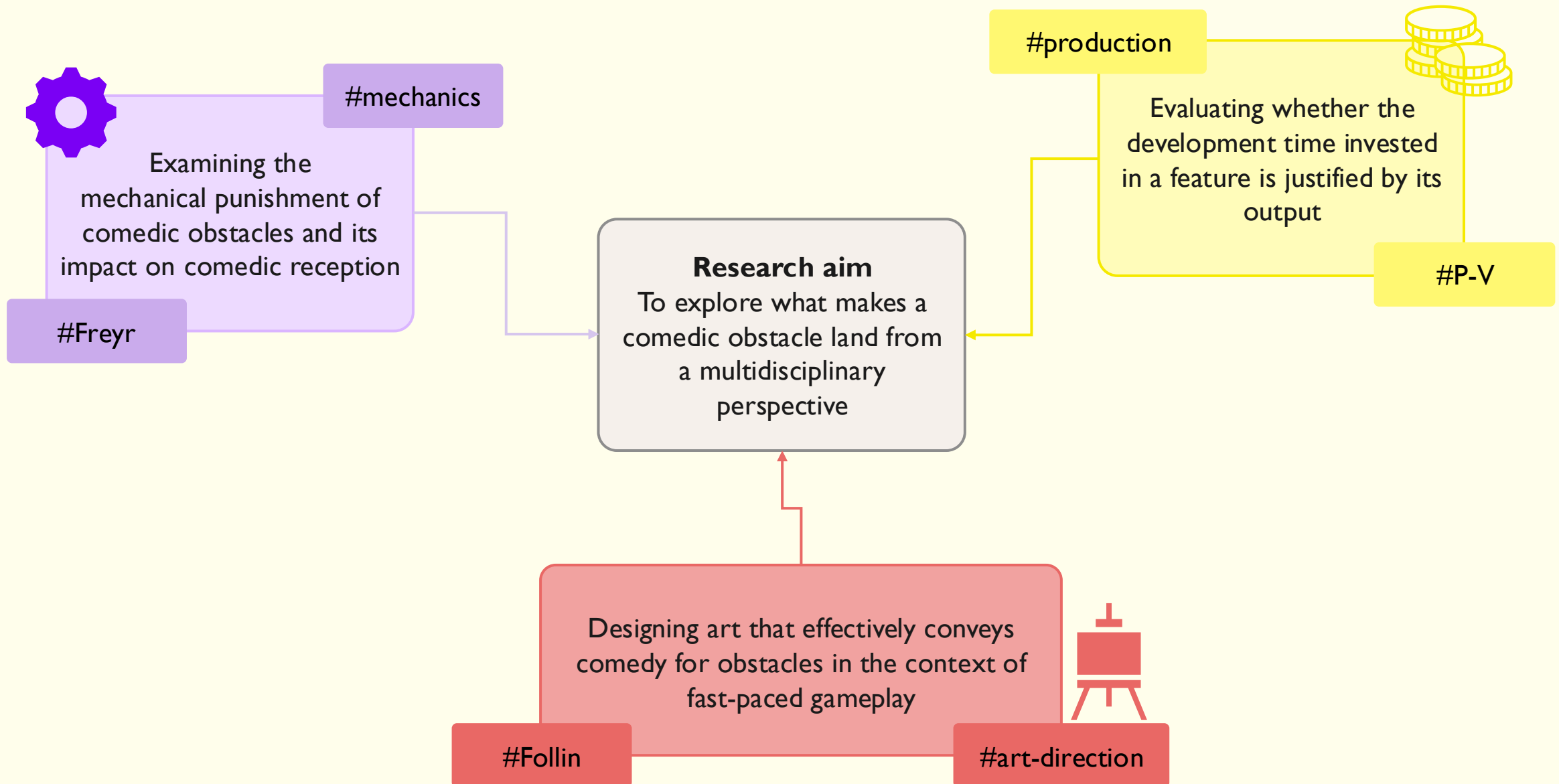
Game Precedents

Theories

Research Gap



Research overview



Game overview

Untitled Driving Game

Steer a car with one stick → control a long arm with another stick → pick up and deliver items along a looping highway → manage impediments

TEAM ROLES

Alexander Freyr

Lead Designer &
Programmer



Alexander Follin

3D & Lighting
Artist



Aleksandra P-V

2D & UI Artist, SFX,
PM



State of the game

Design challenges

- 1st playtest brought low-quality data due to game balancing issues
- Balancing mattered a lot for meaningful data collection

Implemented features

Car traffic → shift-to-shift progress → daytime progression → driving → delivering → spline arm



Trash Bag | Falls from the trash truck → slaps a 2D animal sprite onto the windshield → slows the car

Banana Peel | NPC cars throw it along with harmless trash → spins the player's car



Octopus | Pretends to be human → catches the hand → latches onto the car → steals an item

to the game →

SHIFT



Key findings

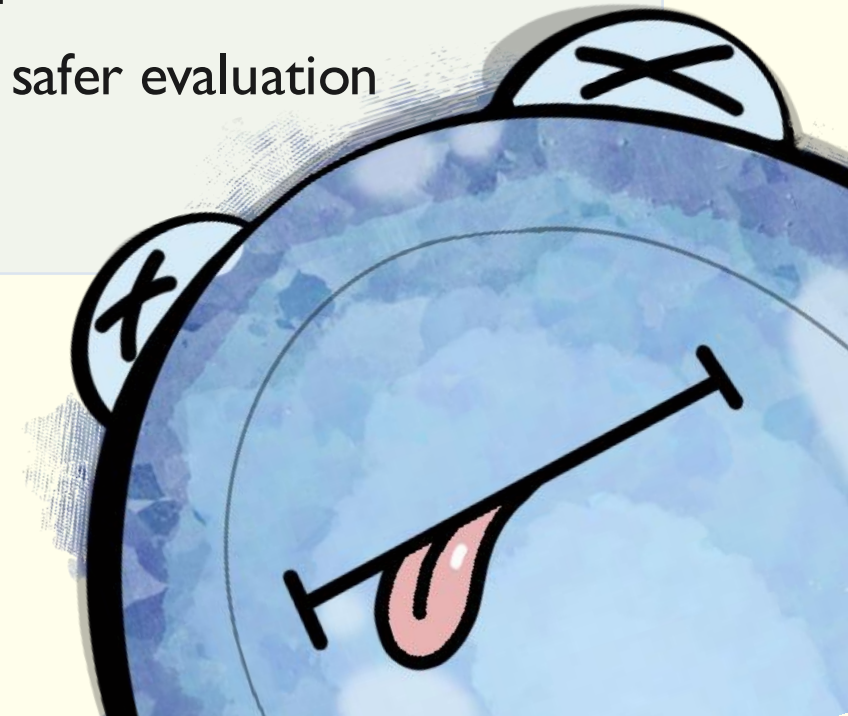
Useful (?) recommendations 😊

(Some) results:

- Art is the most efficient comedic investment
- Behaviour-based comedy is unreliable and expensive
- Comedy can be filtered out as noise if not central to the gameplay in the moment

Research output:

- **Secure the benign** before investing in the violation
- Normalcy is a design resource and can be built up or postponed
- **The three lenses** is a safer evaluation approach



Follin's state of work



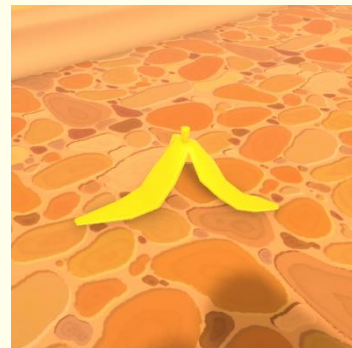
Implemented:

General:

- environment, trash truck, hand, deliverables

Obstacles:

- octopus head, garbage bag, banana peel



Research focus:

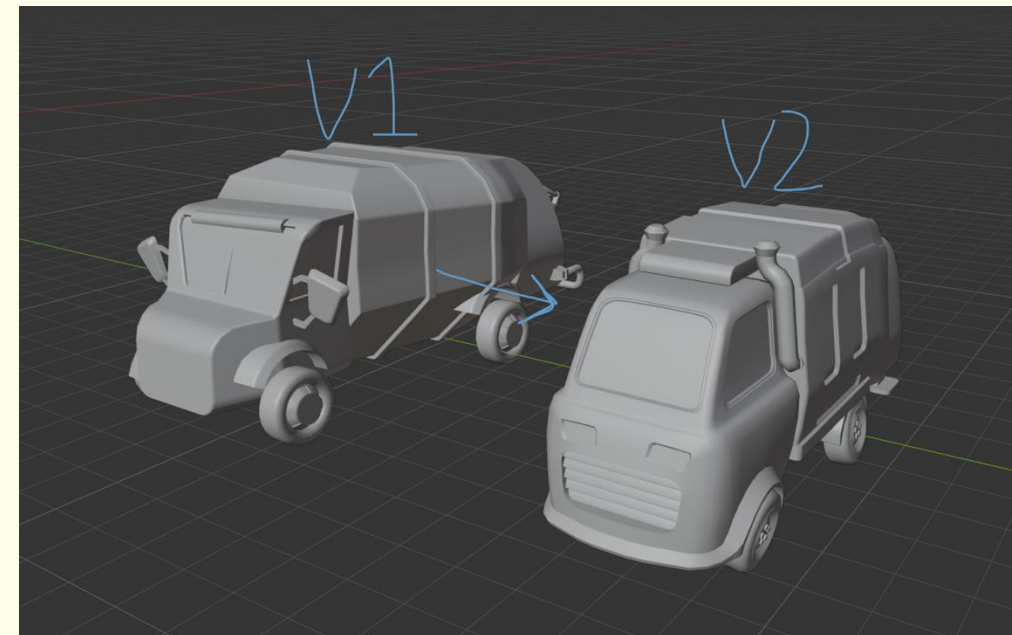
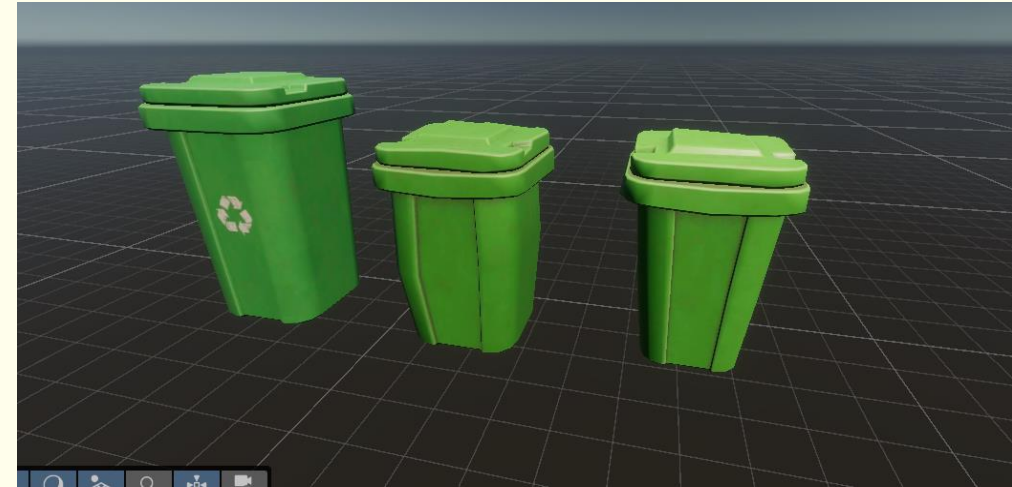
- Render obstacles in a comedic tone, in the context of a fast-paced game.



Follin's Process

Design Workflow:

- Bening Violation Theory
- Fall Guys



Follin's findings



Findings:

1 – Benign violation theory applies heavily, especially the "violation"

2 – Personality and intent plays a key role, likely serves as the "violation"

3 – Element of surprise

Connection to the study:

Providing comedic visuals and general art that forms a cohesive environment in the shape of our decided aesthetic.

Contribution to the field:

Contributes with knowledge about comedy in fast paced game environments and what draws attention, specifically regarding obstacles.

Freyr's state of work



Implemented:

1. Car behavior
2. Arm behavior
3. Obstacle behavior
4. Shift system
5. Delivery system
6. Level system
7. Traffic system

Research focus:

How the mechanical punishment imposed by an obstacle affects the obstacle's comedic reception

Research Process:

1. Conduct interviews, gathering feedback per obstacle
2. Compare & contrast obstacle rating
3. Collecting insightful statements from interviews
4. Analyzing data through the lens of Comedy Theory to produce new findings

Freyr's findings



Findings:

1. Comedy can survive alongside substantial punishment
2. Some disruptions to player agency are read more punitive
3. Mirth inducing obstacles require comedy **during** the punishment
4. Subverting established behavior is effective but may be perceived unintentional

Connection to the study:

My observations of the punishment of comedic obstacles help to explain the relationship between mirth and frustration

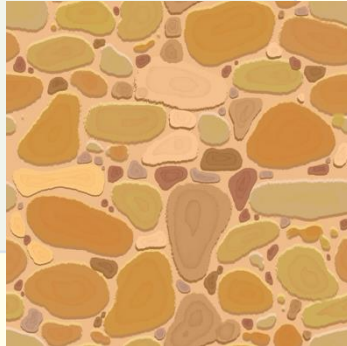
Contribution to the field:

Providing insights into fruitful comedic obstacle design

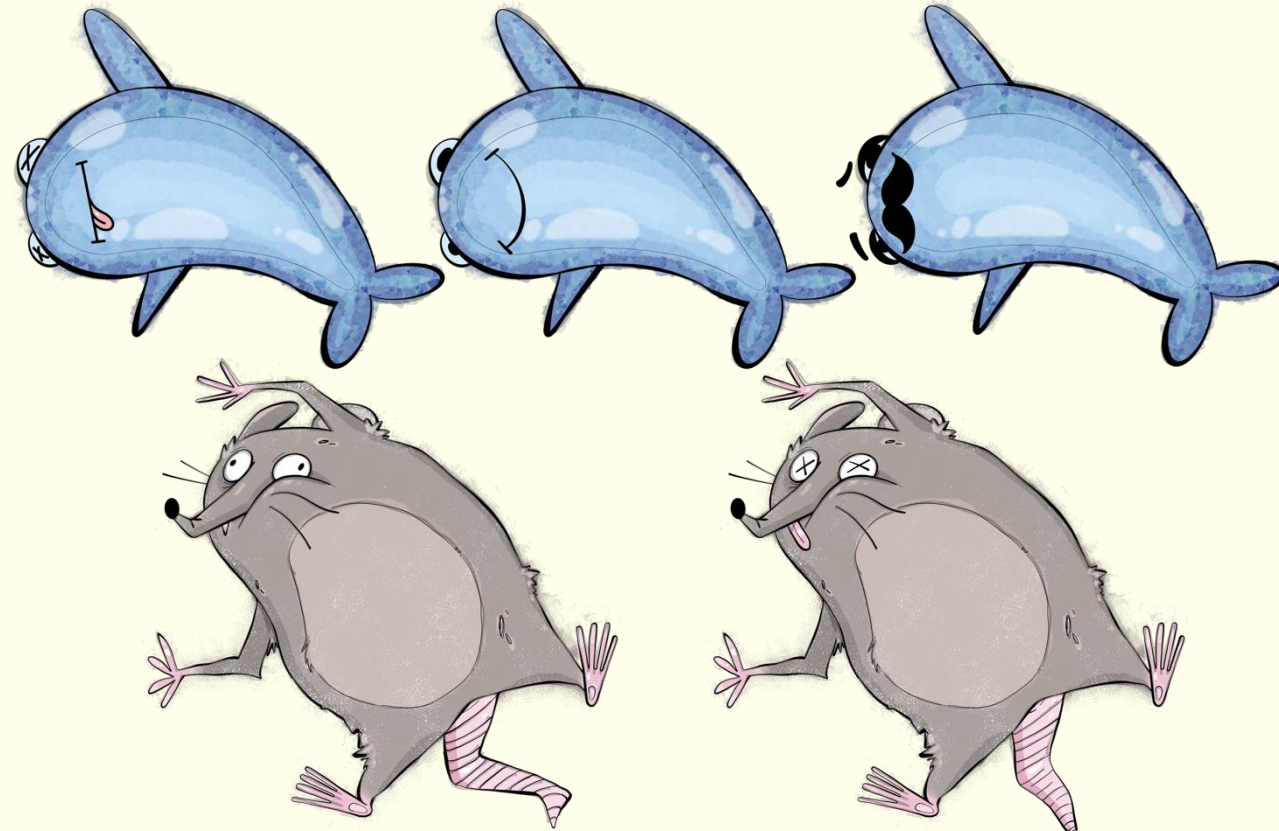
P-V's state of work

Implemented:

- tiling textures for the sky and the road
- 2D sprites for the windshield
- 2D icons for deliveries
- shift UI
- SFX



Research focus: applying Return on Time Investment (ROTI) to the game's impediments to assess the value those bring to the game given the effort



P-V's process

Data: Jira time-tracking + coded playtest responses → ROTI per impediment.

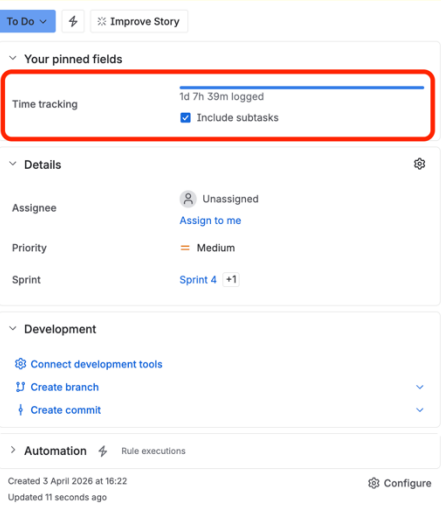
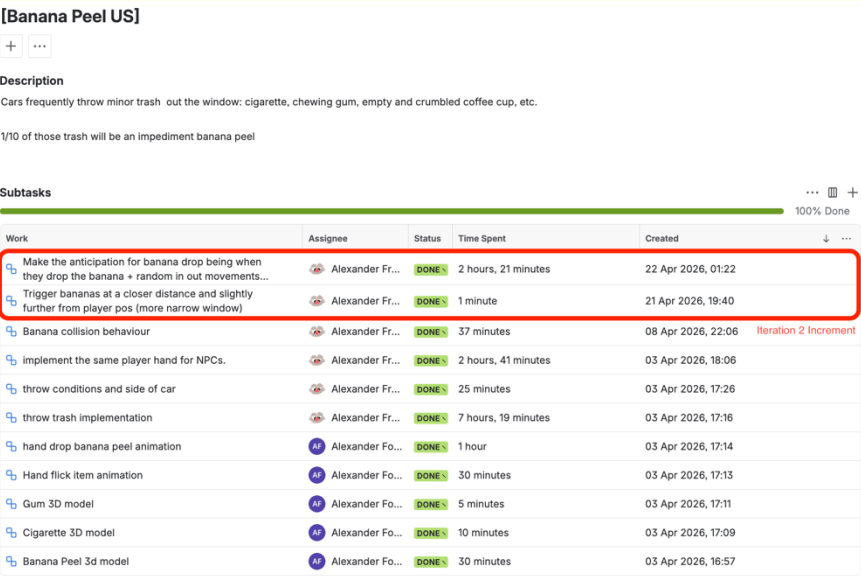


Figure B1

Coding Guide

Code	Lable	Description
+2	Positive comedic	Tester laughed, expressed delight, described the obstacle as funny, surprising, or memorable in a positive way
+1	Mild positive	Tester smiled, found it clever or acceptable, described it as fine or mildly amusing
0	Neutral	Tester had no strong reaction, did not notice the obstacle, or could not remember it
-1	Mild negative	Tester found it slightly annoying, confusing, or disruptive but not strongly negative
-2	Negative	Tester expressed clear frustration, called it unfair, said it broke the flow, or described it as not fun

$$ROTI = \frac{NEV}{NT}, \text{ where } NEV = \frac{EV}{EV_{max}} \text{ and } NT = \frac{t_h}{t_{avg}}$$

12 playtesters*

P-V's findings

Findings:

1. Experience returns and development cost are **poorly correlated** (boom)
2. A design revision can shift experience returns significantly **without proportional time investment**
3. Comedic framing itself *can** function as a **production efficiency multiplier**

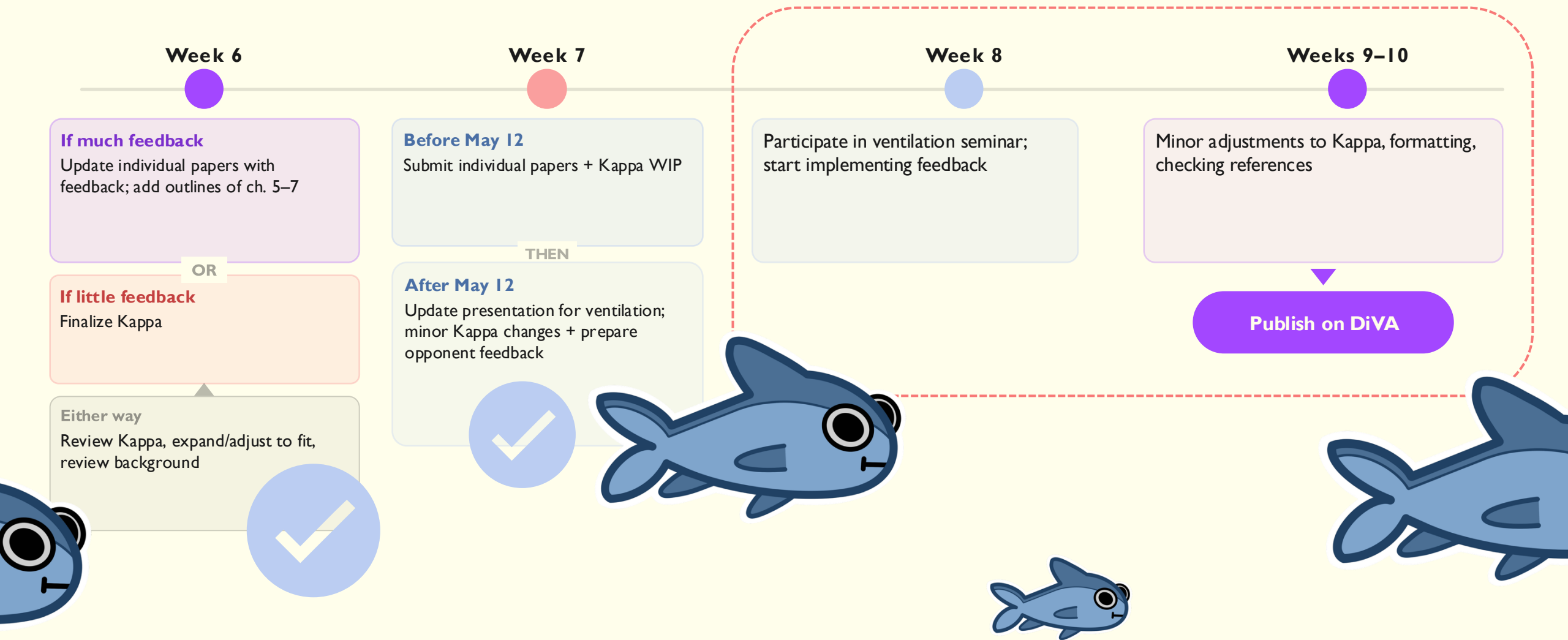
Connection to the study:

- draws from the shared dataset
- outputs a comparable number
- provides resource allocation reasoning

Contribution to the field: a lightweight complementary framework for feature assessment.

→ *Bonus: the formula takes in average production time, so it's applicable to different teams.*

Planning forward



Questions time!

THESIS | DEVELOPMENT

Alexander Freyr | Alexander Follin | Aleksandra Pavlova-Verevkina

